



Writing small numbers in standard form

Task A: Writing very small numbers in standard form

1) Using the place value charts, write the following in standard form.

a)

O	t	h	th	tth	hth
10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
0	0	0	0	6	9

b)

O	t	h	th	tth	hth
10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
0	0	1	2	8	

2) Which of the following are not in standard form? Explain why.

a) 123×10^{-9}

b) 1.24×10^{-3}

c) 0.96×10^{-1}

d) $12\,300 \div 10^{10}$

e) 0.059×10^8

3) Without a place value chart, write the following in standard form.

a) 0.000 000 354

b) 0.001 45

c) 0.000 000 000 563

d) 0.048 6

e) 0.000 000 000 631



Task B: Converting between ordinary and standard form

1) Convert the following into standard form.

- a) 0.000 000 263
- b) 652 000 000
- c) 0.005 63
- d) 892 000 000 000
- e) 0.000 000 000 000 0003
- f) 30 004 000
- g) 0.000 300 4

2) Convert the following into ordinary numbers

- a) 2.31×10^4
- b) 6.153×10^{-5}
- c) 9.1565×10^{-6}
- d) 2.48×10^5
- e) 2.48×10^{-5}
- f) 3.0065×10^{-9}

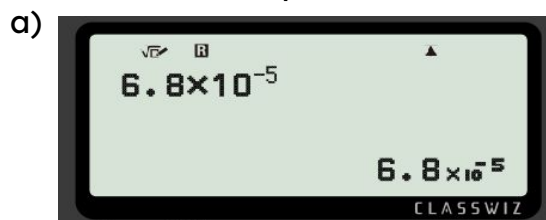
3) A microchip has a width of 2×10^{-6} m. Write this as an ordinary number in cm.

4) Mars is approximately 225 million km away from Earth. Write this in standard form in m.

5) The radius of an atom is approximately 1×10^{-10} m. Write this as an ordinary number in mm.

Task C: Standard form on the calculator

1) Write the output as a decimal.



2) Investigate when the number becomes so small that the calculator converts it into standard form.

3) Fill in the missing information.

